Analytical Modelling of Power Dissipation and Voltage Swing of CMOS Logic Circuit for Near-Threshold Computing

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***Abstract*—In near threshold computing (NTC), by aggressively scaling down the supply voltage to values near the threshold voltage, considerable reduction in power consumption and improvement in energy efficiency have been observed recently. In contrast to these advantages, there are problems like logic failures, performance variation that comes with NTC approach. Ambient temperature has been found to affect performance as well. In this paper,analytic expression of the power dissipation and the noise margin of a Inverter in near threshold regime have been done. Analysis of 1-bit, 4-bit, 8-bit and 16-bit Carry Look Ahead adders (CLAs) have been done in the near-threshold regime to show the workability of the near threshold logic circuits. Extensive CADENCE simulations have been done to verify the analytical modeling.**

Keywords—near-threshold; power; energy-efficiency; CMOS; CLA;

I. INTRODUCTION

Power and energy efficiency have become an unavoidable design constraints with the advent of ultra-low power and energy-harvested systems [1]-[4]. Mainly in portable applications, sensor nodes, Internet on Things (IoT). applications, biomedical applications etc ultra low power consumption is the pivotal issues instead of the performance. Recently, sub-threshold operation [1]-[4], supply voltage has been scaled down below the threshold voltage of the MOSFET to meet the requirement of ultra low power consumptions. Hence leakage current functions as the main current which is not capable of driving high loads. Moreover, sub-threshold operations are very much sensitive to the process parameter variations and offers very poor noise margin.

In contrast to the sub-threshold operations, near threshold computing (NTC) [5]-[8] offers higher energy efficacy with an improved performances. In NTC, the supply voltage has to set greater than the threshold voltage but not so much higher like the conventional operation. For example, in 45 nm technology node, the supply voltage will be approximately within 400 mV to 500 mV compared to the 1 V supply voltage of conventional operations [5], [9], [10]-[11]. This gives a wider scope to develop newer low-voltage semiconductor devices with a bit of improvement in the performances.However, NTC is a good fit for developing sensor-based systems for biomedical applications, image processing, also caters to graphics performance and high performance computing (HPC). NTC gives a greater throughput and energy efficiency while compromising on frequency and die area.

In spite of higher energy efficiencies and improvement in performance, reliability sometimes imposes limitations in the Near-Threshold Computing (NTC). Also, variation in temperature can affect the performances sometimes. So, in depth analytical study is necessary in NTC to address the issues and to propose a suitable way to mitigate the challenges. In literature, few architectures have been proposed for NTC. However, the reported architectures did not present the analytical details of the different issues for better understandings. In this paper, for the first time in literature, we have detailed the analytical expression of the power dissipationsand the noise margin, verified the proposed models through extensive CADNECE simulations. The analysis will be useful for the better understanding of the challenges of NTC and can be mitigated.

II. NEAR-THRESHOLD CMOS LOGIC

A. Basic Model of MOSFET in Near-Threshold Regime

The I–Vcharacteristics of the near-threshold PMOS device can be expressed by

$$I_P = K_P\left(\frac{W}{L}\right)_P [(V_{SG} - |V_T|)]V_{SD} - \frac{1}{2}V_{SD}^2] \quad (1)$$

Where,K_P is the transconductance of PMOS; W/L is the aspect ratio of PMOS; V_{SG}, V_{SD}, and V_T are source to gate, source to drain, and threshold voltage of PMOS, respectively. In near-threshold regime, the supply voltage (V_{DD}) is set to

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almost two times of the threshold voltage (V_T). In 22 nm, the V_{DD} is chosen as 500 mV which is almost twice of the threshold voltage of the NMOS transistors. Like the super-threshold regime, when a logic "0" is applied at the gate of PMOS, it will turn on and when a logic "1" is applied at the gate, it will turn off, provided the source and drains are connected with the V_{DD} and output respectively.

When, PMOS is turned "ON", $V_{SD} = \Delta V$. So from (1),

$$I_P = K_P \left(\frac{W}{L}\right)_P [V_{DD} - |V_T|]\Delta V \tag{2}$$

Therefore,the value of resistance offered by the PMOS is given by,

$$R = \frac{\Delta V}{I_P} = \frac{1}{K_P \left(\frac{W}{L}\right)_P [V_{DD} - |V_T|]} \tag{3}$$

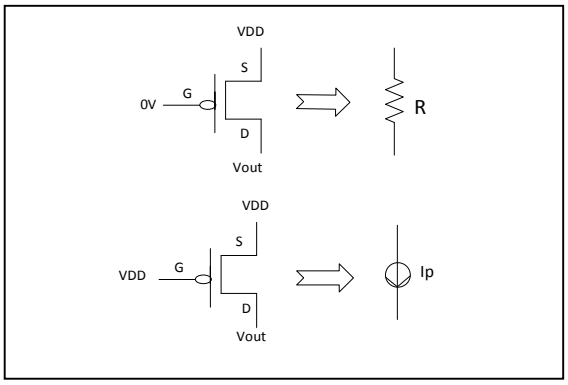


Fig. 1.Equivalence of the MOSFET in near-threshold regime.

So, when a PMOS is "ON", it can be represented as a resistor in near threshold regime. Similarly, when the PMOS is "OFF", leakage current will flow through the PMOS and can be represented by a constant current source of negligible value. From the EKV model [10], the drain current of an NMOS can be expressed as,

$$I_P = \beta_P \exp\left(\frac{V_{SG}}{n_P V_T}\right)\left[1 - \exp\left(\frac{-V_{SD}}{V_T}\right)\right] \tag{4}$$

Hence, V_T is the thermal voltage (=KT/q); n_P is the sub-threshold slope of PMOS. β_P can be used to measure the transistor strength and depends on the process parameter [2].

For low gate input, $\exp\left(\frac{V_{SG}}{n_P V_T}\right)$=1 as V_{SG}=0 and $\exp\left(\frac{-V_{SD}}{V_T}\right)$=0 as $V_{SD} >> V_T$. So, (4) can be simplified as,

$$I_P \approx \beta_P \tag{5}$$

The equivalence of the PMOS in near threshold regime has been given in Fig. 1.

B. *Energy Dessipation of CMOS inverter in Near-Threshold Regime*

In this section, the analytical expression of power dissipation and the noise margin of near threshold logic circuits have been detailed. Simple inverter has been adopted as a reference circuit. A load capacitor (C_L) has been connected with the output of the inverter to provide stable output waveforms. Considering the output voltage (V_{out}) across the load capacitor, the voltage drop (ΔV) across the PMOS ($\approx R_P$) during charging can be represented as,

$$C_L \frac{\partial V_{out}}{\partial t} = \frac{V_{DD} - V_{OH}}{R_P} = \frac{\Delta V}{R_P} \tag{6}$$

Hence, V_{OH} and V_{DD} are the output high voltage during the charging phase of the inverter and the supply voltage respectively. V_{OH} can be obtained by solving (6) as follows,

$$R_P C_L \int_0^{V_{OH}} \frac{dV_{OUT}}{V_{DD} - V_{OUT}} = \int_0^T dt \tag{7}$$

$$V_{OH} = V_{DD}\left[1 - e^{\left(\frac{-T}{R_P C_L}\right)}\right] \tag{8}$$

Substituting the expression of V_{OH} in (6),we get

$$\Delta V = V_{DD} e^{\left(\frac{-T}{R_P C_L}\right)} \tag{9}$$

if $\frac{T}{R_P C_L} >> 1$, then $\Delta V \approx 0$. Hence, the frequency of operation for the NTC can be obtained easily on basis of the relationship ($\frac{T}{R_P C_L} >> 1$). The operating frequency (f_{op}) must be,

$$f_{op} << \frac{1}{R_P C_L} \tag{10}$$

The expression of V_{OL} can be given as,

$$V_{OL} = \Delta V = V_{DD} e^{\left(\frac{-T}{R_P C_L}\right)} \tag{11}$$

The equation of voltage swing for CMOS inverter can be given as,

$$V_{swing} = V_{OH} - V_{OL} \tag{12}$$

$$= V_{DD}\left[1 - 2e^{\left(\frac{-2T}{R_P C_L}\right)}\right]$$

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V_{OH} is the high output voltage and V_{OL} is the low output voltage.

The energy dissipationin the above-mentioned near-threshold CMOS based inverter during the charging process can be calculated as

$$E = \int_0^T \frac{\Delta V^2}{R} dt \tag{13}$$

Now,substituting the expression of ΔV from (9) in (13) equation (5), the energy equation comes to be

$$E = \frac{V_{DD}^2}{R} T \left[e^{\left(\frac{-2T}{R_P C_L} \right)} \right] \tag{14}$$

Hence also, the energy dissipation would very much smaller when the operating frequency will be, $f_{op} << \frac{1}{R_P C_L}$.

III. Design and Analysis of Complex Circuits in Near – Threshold Regime

In this section, design and analysis of near threshold CMOS logic-based 1-bit,4-bit,8-bit and 16-bit CLA are given to shown.The basic block diagram is implemented using simple gates like NAND,NOR,XOR.The digital gates of the library are developed at transistor level. Hence, 22-nm technology file is used in our transistor-level designs .The basic building block of 1-bit CLA is given in Fig.2. The expression of the *i* th sumand and the (*i* + 1)th carry output can be given as

$$S_i = A_i \oplus B_i \oplus C_i$$
$$C_{i+1} = A_i B_i \oplus C_i (A_i \oplus B_i) \tag{15}$$

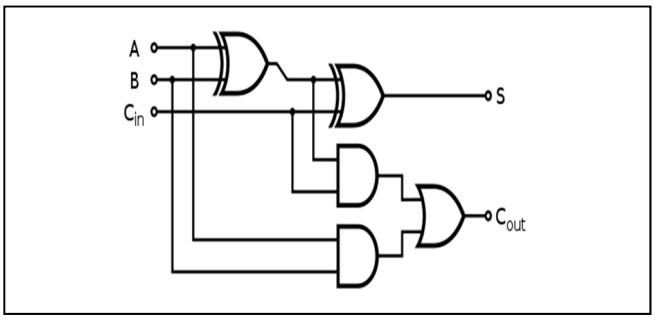


Fig.2. Basic building block of 1-bit CLA.

The 4-bit,8-bit and 16-bit CLA's are designed by cascading this 1-bit CLA.Simulated output waveform, given in Fig.3. Hence, in 1-bitCLA ,the sum and carry outputs i.e.,S0 and COUT0 are given to show a better understanding and workability of the proposed logic. During the simulation, we set arbitrary values of A and B along with Cin.The values are A0 = 0000, B0= 1111 and Cin =0000.Therefore,the output S0=1111 and COUT0=0000 as no carry is generated.Similarly,for 4-bit, given in Fig.4.,four sum outputs, i.e., S0, S1, S2, and S3 along with the final carry output, COUT3, are given in Fig. to show a better understanding andworkability of the proposed logic. Hence, the inputs and outputs, i.e., A, B, and S are represented as, A = A3A2A1A0, B = B3B2B1B0, and S = S3S2S1S0, respectively. During the simulation, arbitrary values of A and B along with COUT0 are set.The values are A0 = 0000, A1 = 0000, A2 = 0000,A3= 0000, B0 = 1111, B1 = 1111, B2 = 1111, B3 = 1111, and Cin= 0000. Initially, carry is set to logic 0. The output will be S0 = 1111; S1 = 1111; S2 = 1111; and S3 = 1111. The carry output will be COUT3= 0000, as no carry will be generated.The same is done in case of 8-bit and 16-bit.

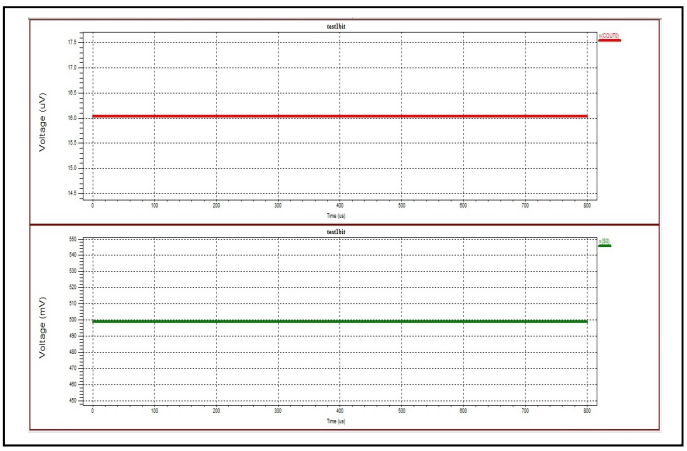


Fig.3. Simulation of 1- bit adder in the near threshold regime.

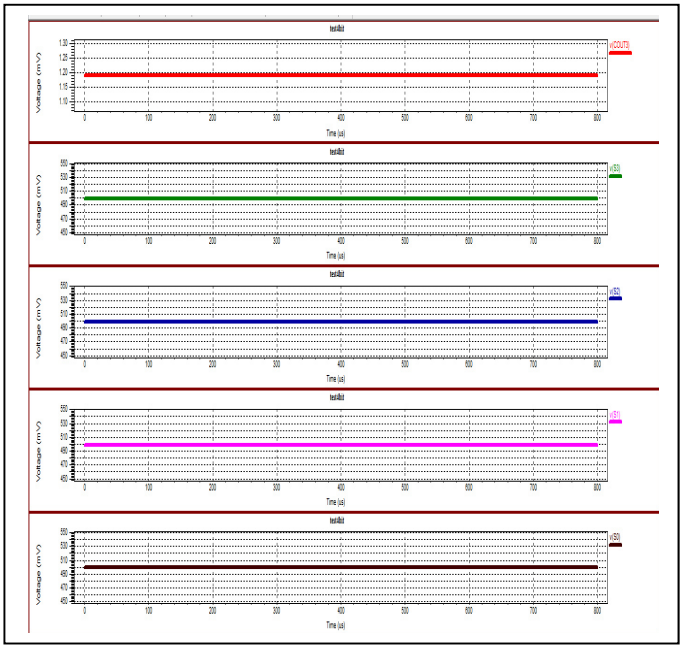


Fig.4. Simulation of 4- bit adder in the near threshold regime.

Fig.5. gives the variation of voltage swing and average power of 1-bit,4-bit,8-bit and 16-bit CLA with variation in the supply voltage. It is observed that with increase or decrease in supply

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voltage, the voltage swing changes linearly. The voltage swing of the adders remains almost equal to the supply voltages.For 1-bit CLA ,the voltage swings at supply voltage 375mV and 500mV is 374.19mV and 499.08mV respectively. Similarly, for 4-bit and 16-bit ,it is almost equal to the voltage supply. But in case of 16-bit CLA, the voltage swing is 350.74mV and 456.93mV at 375mV and 500mV respectively. A small drop of 6% to 8% can be seen in case of 16-bit and this is due to the fact that the number of pMOS and nMOS transistors in 16-bit is very high.

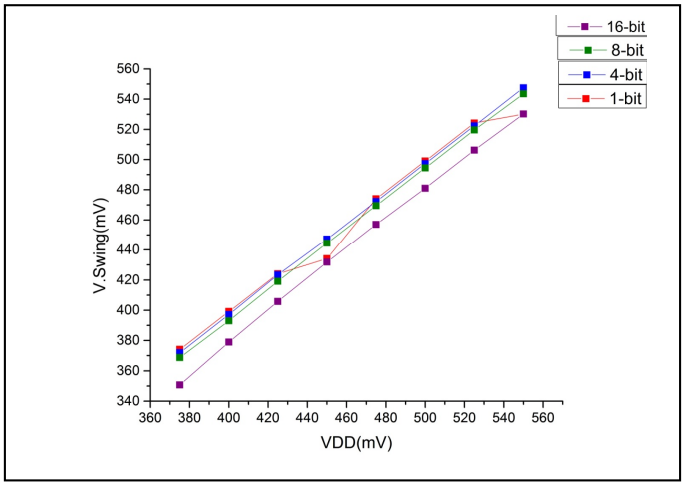


Fig.5. Effect of voltage supply variation on voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA

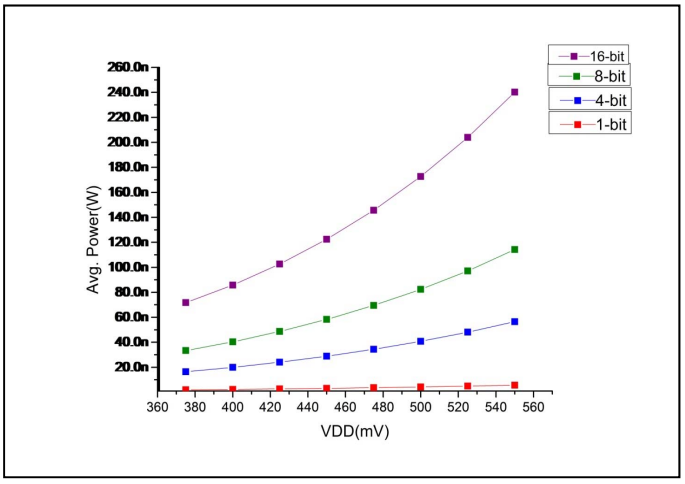


Fig.6. Effect of voltage supply variation on power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA.

In Fig.6.it is observed that the power dissipation of near-threshold CMOS-based 1-bit,4-bit,8-bit and 16-bit CLA increases with the increase in the supply voltage.We can see that with variation in voltage supply, near-threshold current will vary linearly.Therefore, output node swing can be affected due to this.This graph gives the comparison between the power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA with variation in voltage supply.It is observed that the power dissipation is maximum and in the range of 70nW to 240nW in case of 16-bit CLA .Whereas,it is minimum and in the range of 0nW to 10nW in case of 1-bit CLA.

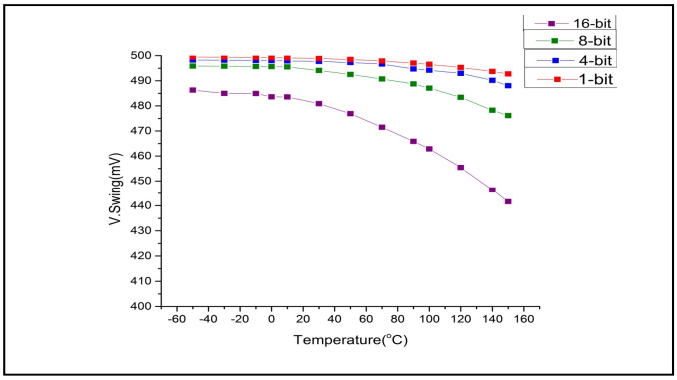


Fig.7.Effect of temperature variation on voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA.

Fig.7., gives the variation of voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA with rise in temperature.It is observed that with increase in temperature,the voltage swing decreases.The voltage swing of the 16-bit adder falls more rapidly than the other three.This is due to the fact that the number of pMOS and nMOS transistors in 16-bit is very high. At 500mV supply voltage and 0°C, the voltage swing for 1-bit,4-bit,8-bit and 16-bit CLA is 499.23mV,498.03mV ,495.63mV and 483.63mV respectively. At room temperature(30°C), the voltage swings are 498.87mV,497.68mV,494.08mV and 480.89mV for 1-bit,4-bit,8-bit and 16-bit CLA respectively.So,it can be seen that there is more than 3% degradation in case of 16-bit CLA and less than 1% in 1-bit CLA. In addition, the power dissipation due to leakage has a direct dependency with temperature.

In Fig8, it is observed that the power dissipation of near-threshold CMOS-based 1-bit,4-bit,8-bit and 16-bit CLA increases with the increase in temperature. It can be seen that in case of temperature variations, near-threshold current will vary as it depends linearly on temperature.

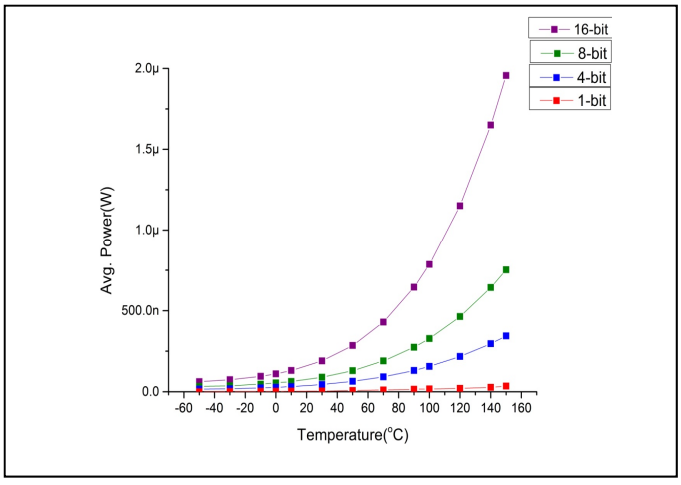


Fig.8.Effect of temperature variation on power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA.

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Therefore, output node swing can be affected due to the temperature variation .This graph gives the comparison between the power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA with varying temperatures.It is observed that the power dissipation is maximum in case of 16-bit CLA and minimum in case of 1-bit CLA.The power dissipation of 16-bit CLA reaches the range of 2 microWatt while other remains in the range of nanoWatt.

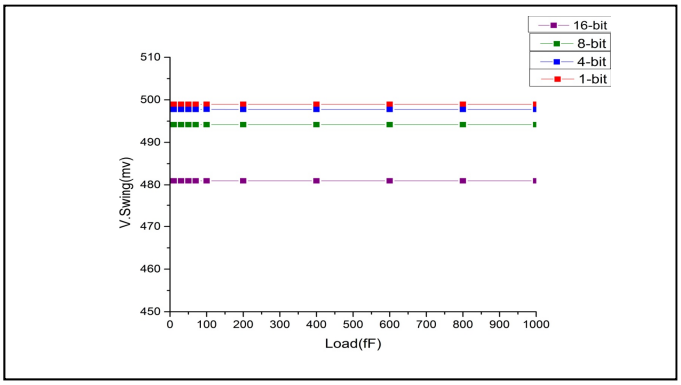


Fig.9.Effect of load variation on voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA.

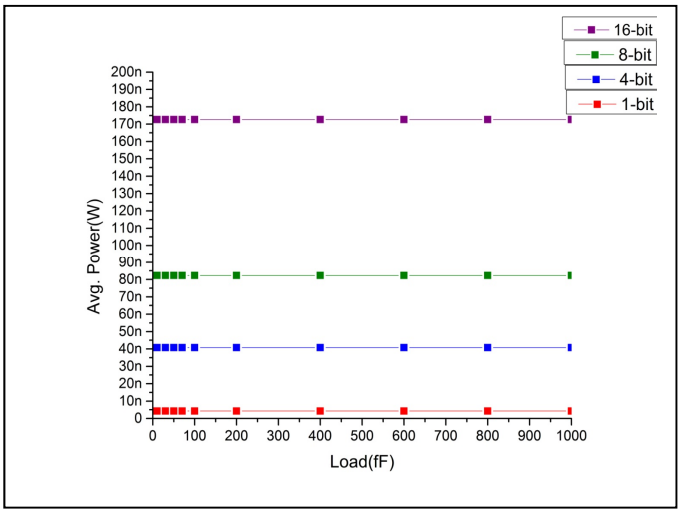


Fig.10. Effect of load variation on power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA.

Fig. 9 and Fig. 10 give the variation of voltage swing and average power of 1-bit,4-bit,8-bit and 16-bit CLA with variation in the output load. Both the graphs gives the comparison between the voltage supply and power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA with variation in output load.We can see that with increase in load,the voltage swing does not change. For 1-bit,it remains constant at 498.95mV,for 4-bit 497.75mV,for 8-bit 494.16mV and for 16-bit 480.96mV.Since,there is no change in voltage swing with variation in load,the average power dissipation also remains constant. For 1-bit ,it remains constant at 5nW,for 4-bit 40nW,for 8-bit 80nW and for 16-bit 170nW.It is clear that average power dissipation is maximum in case of 16-bit adder and minimum in case of 1-bit adder.

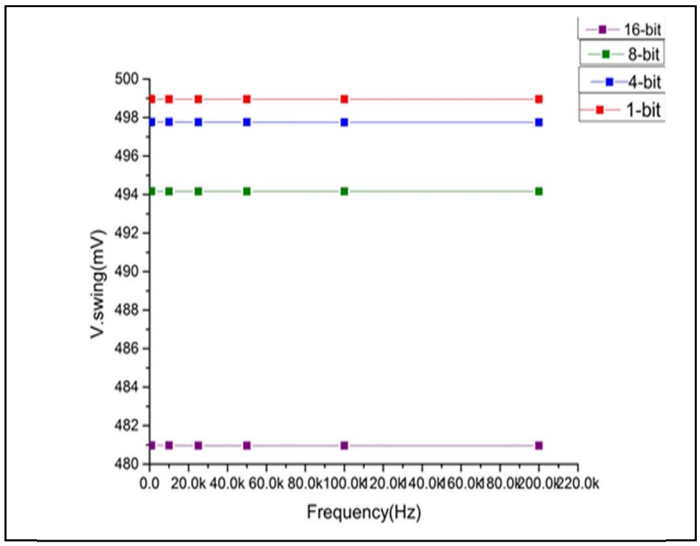


Fig.11.Effect of frequency variation on voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA.

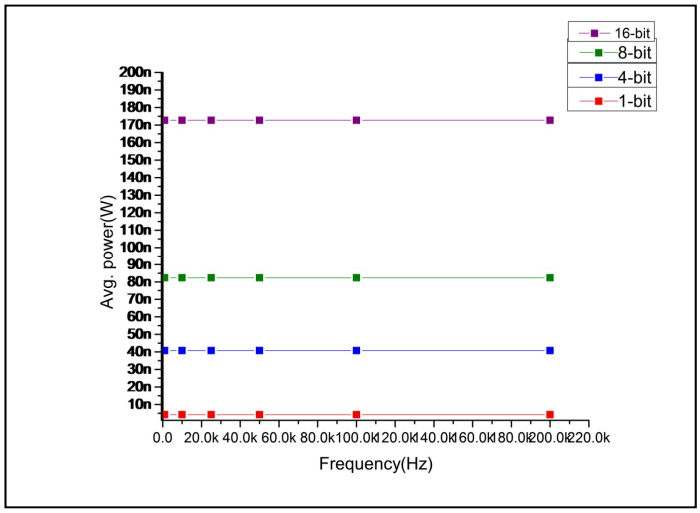


Fig.12. Effect of frequency variation on power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA

Fig.11. and Fig.12.give the variation of voltage swing and average power of 1-bit,4-bit,8-bit and 16-bit CLA with variation in frequency. Both the graphs gives the comparison between the voltage supply and power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA with variation in frequency.Frequency is increased upto 2MHz but we can see that with increase in frequency,the voltage swing does not change. For 1-bit ,it remains constant at 498.95mV,for 4-bit 497.75mV,for 8-bit 494.16mV and for 16-bit 480.96mV .It remains parallel to the x-axis.Since,there is no change in voltage swing with variation,the average power dissipation also remains constant. Also it is observed that the average power dissipation is maximum in case of 16-bit adder and minimum in case of 1-bit adder.

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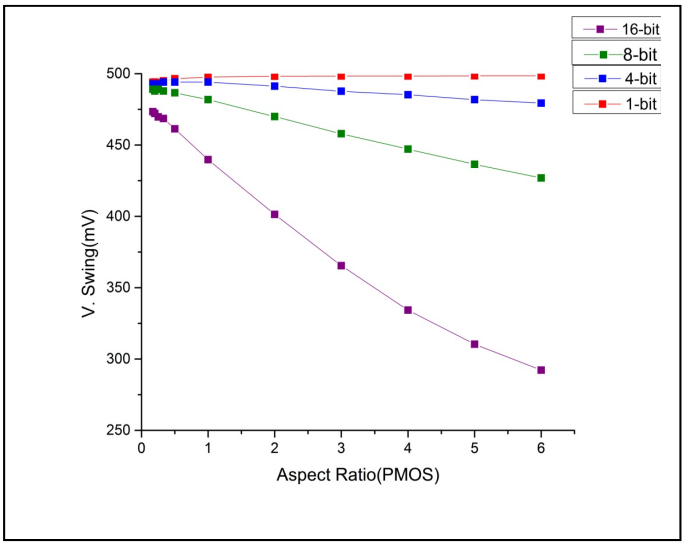


Fig.13.Effect of variation in aspect ratio on voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA.

The above Fig.13.gives the variation of voltage swing of 1-bit,4-bit,8-bit and 16-bit CLA with variation in the aspect ratio of pMOS.The aspect ratio of nMOS is kept fixed(W=88nm amd L=44nm).Aspect ratio of pMOS is varied from 1/6 to 6.It is observed that with increase in aspect ratio,the voltage swing decreases.Therefore,voltage swing is inversely proportional to the aspect ratio of pMOS. The voltage swing of 1-bit and 4-bit CLA almost remains equal to supply voltage.But the voltage swing of 8-bit CLA degrades by 15% and for 16-bit CLA,it is highly degraded when aspect ratio is 6.

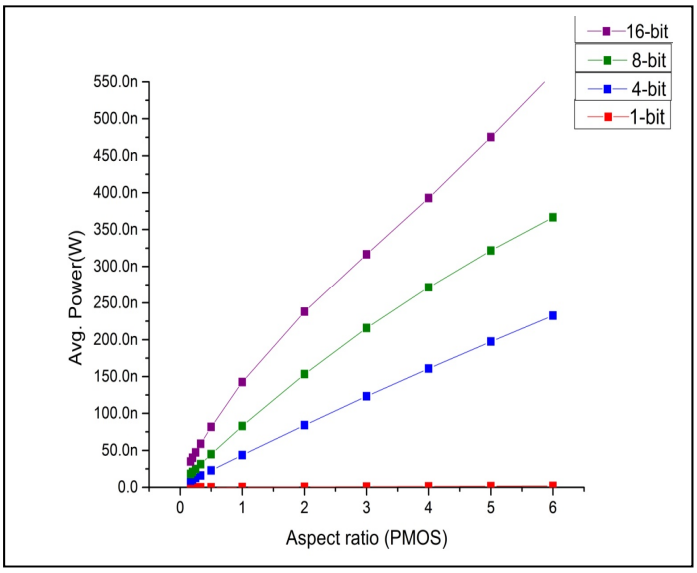


Fig.14. Effect of aspect ratio variation on power dissipation of 1-bit,4-bit,8-bit and 16-bit CLA.

In Fig.14.we observe that the power dissipation of near-threshold CMOS-based 1-bit,4-bit,8-bit and 16-bit CLA increases linearly with the increase in aspect ratio of pMOS. It is maximum for 16-bit and minimum for 1-bit.For 1-bit,average power dissipation is less than 5nW while for 16-bit CLA,it increases upto 500nW when aspect ratio is greater than 5.

IV. Conclusion

The analysis of CLAs in the near-threshold regime gives a fair amount of information regarding the effect of various circuit metrics, like supply voltage variation, aspect ratio variation, frequency , temperature, load , on voltage swing and power dissipation. The comparison among 1-bit, 4-bit, 8-bit and 16-bit CLAs show the trends in behavior of systems with increasing circuit complexity. This will definitely help in developing complex yet power -efficient, functional systems using CMOS logic for ultra-low power applications. Near-Threshold Computing (NTC) is a promising field and the work shown in this paper is but a step towards incorporating NTC in the development of energy-efficient systems utilizing CMOS logic.

References

[1] A.Wang, B.Calhoun, and A.Chandrakasan,"Subthreshold Design for Ultra Low-Power Systems," New York: Springer, 2006.

[2] M. E. Sinangil, N. Verma, A. P. Chandrakasan, "A reconfigurable 65nm SRAM achieving voltage scalability from 0.25-1.2V and performance scalability from 20kHz-200MHz",*ESSCIRC*, 2008.

[3] H. N. Patel, F. B. Yahya, B. H. Calhoun, "Optimizing SRAM Bitcell Reliability and Energy for IoT Applications", *ISQED*, 2016.

[4] M.Alioto,"Understanding DC Behavior of Subthreshold CMOS Logic through Closed-Form Analysis," IEEE J. Circuits and Systems,vol.57 , no.7, pp. 1597-1607, July 2010.A.

[5] R. G. Dreslinski et al., "Centip3De: A 64-Core, 3D Stacked Near-Threshold System", IEEE Micro, vol. 33, issue 2, 2013.

[6] D. Fick, "Centip3De: A 3930 DMIPS/W Configurable Near-Threshold 3D Stacked System with 64 ARM Cortex-M3 Cores", *Proc. IEEE Int'l Solid-State Circuits Conf.*, pp. 190-192, 2012.

[7] B. Zhai, "Energy-Efficient Near-Threshold Chip Multi-Processing", *Proc. ACM/IEEE Int'l Symp. Low Power Electronics and Design*, pp. 32-37, 2007.

[8] R. G. Dreslinski et al., "Near-threshold computing: Reclaiming Moore's law through energy efficient integrated circuits," Proc. IEEE, vol. 98, no. 2, pp. 253–266, Feb. 2010.

[9] M. R. Kakoee, L. Benini, "Robust Near-Threshold Design With Fine-Grained Performance Tunability", IEEE TCAS-I, vol. 59, no. 8, pp. 1815-1825, Aug. 2012.

[10] D. Markovic et al., "Ultralow-power design in near-threshold region," Proc. IEEE, vol. 98, no. 2, pp. 237–252, Feb. 2010.

[11] Y.Tsividis, Operation and Modeling of theTransistor MOS , 2nded.Oxford,U.K.:OxfordUniversityPress,2003.

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